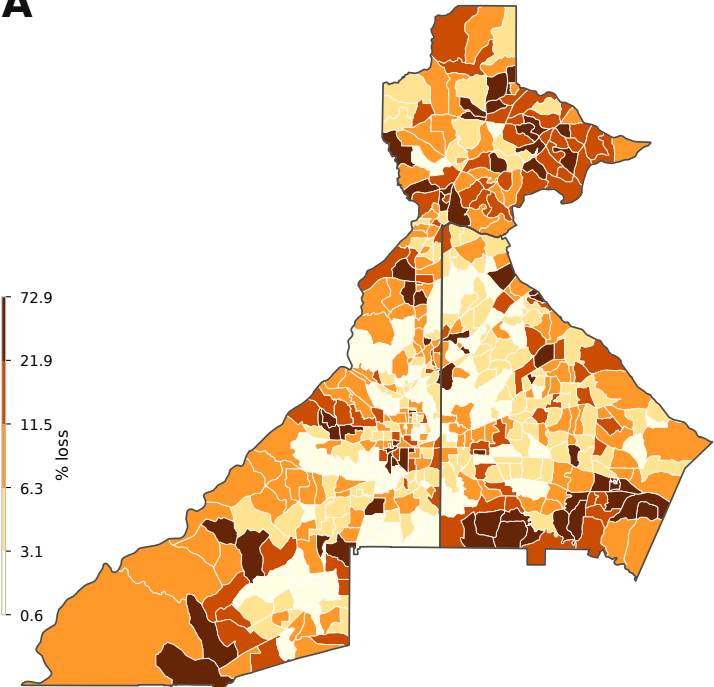


CartoLLM Autonomous Validation · Atlanta Metro Census Tracts (Fulton + DeKalb Counties, GA · Real TIGER Geometry)

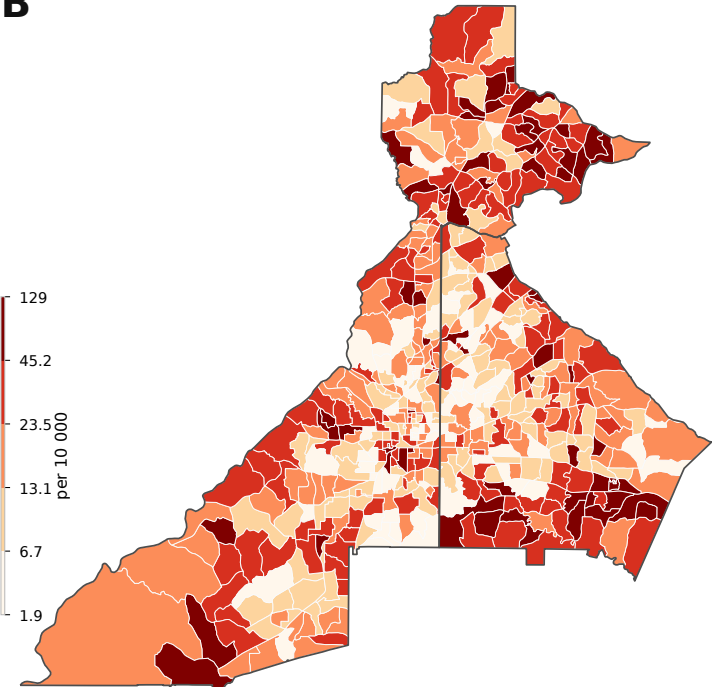
A

Tree Canopy Loss (%)



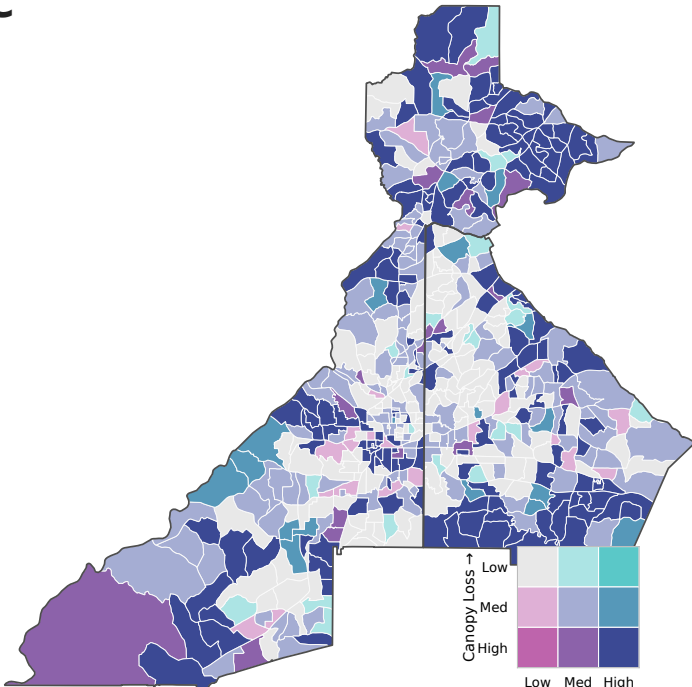
B

Asthma Hospitalisation Rate (per 10 000)



C

Bivariate: Canopy Loss × Asthma Rate



Gate 2 — Classification Diagnostic Engine

Variable	Diagnosis	Prescribed	Status
Tree Canopy Loss	heavy right skew	log transform then jenks	PRESCRIBED
Asthma Hosp. Rate	heavy right skew	log transform then jenks	PRESCRIBED

LLM proposed naive Jenks. Gate 2 overrode both variables to log-t
ransform + Jenks on transformed residual distributions.

Gate 3b — Bivariate Spatial Cross-Correlation

Statistic	Value	p-value	Threshold	Decision
Bivariate Moran's I_xy	+0.3262	0.0050	I_xy > 0.15	✓ PASS
Spearman's rho	+0.9471	0.00e+00	rho > 0.20	✓ PASS

✓ Overall gate decision: APPROVE

199 permutations under H₀ of no spatial cross-association. Pseudo p-value = (M+1)/(R+1).